

Progress across our 15 priority materials



Aluminum

In 2024, approximately 71 percent of the aluminum in products we shipped to customers came from recycled sources.³¹ Learn more on [page 14](#).





Cobalt

In 2024, 76 percent of the cobalt shipped in our products — up from 52 percent in 2023 — came from certified recycled sources using mass balance allocation.





Copper

In 2024, we achieved 100 percent recycled copper in multiple printed circuit boards across products.³² We also introduced 100 percent recycled copper in multiple thermal module components of Mac mini, 16-inch MacBook Pro, and iMac (four ports).³³





Glass

We expanded our use of recycled glass to select components of Apple Watch Series 10 and iPad mini.





Gold

In 2024, our use of recycled gold increased to about 40 percent across all product lines — up from 1 percent in 2021.





Lithium

In 2024, approximately 53 percent of the lithium shipped in our batteries came from certified recycled sources using mass balance allocation, including postindustrial scrap and postconsumer scrap from end-of-life batteries — up from 24 percent in 2023.





Paper

For information about our progress with paper, see our [feature](#) on packaging.



Plastics

In 2024, we used at least 50 percent recycled plastic in 20 components of iPhone 16 and iPhone 16 Plus. We also used 25 percent recycled plastic in the antenna lines, made from upcycled plastic bottles.





Rare earth elements

In 2024, more than 80 percent of the total rare earth elements that we shipped in products came from certified recycled sources — up from 75 percent in 2023.





Steel

In 2024, we introduced the use of 80 percent recycled steel in the speaker attachment and Taptic Engine in Apple Watch Series 10.





Tantalum

We continue our efforts to source 100 percent recycled tantalum and are actively investigating recovery solutions from end-of-life electronics to expand the availability of recycled tantalum.





Tin

In 2024, we used more than 45 percent recycled tin on average across all product lines — up from 40 percent in 2023.





Titanium

In 2024, Apple Watch Series 10 contained 95 percent recycled titanium in the case.





Tungsten

In 2024, approximately 99 percent of the tungsten used came from recycled sources. Our disassembly robot, Daisy, and our recycling machine, Dave, helped recover and recycle tungsten from the Taptic Engine in our products.





Zinc

Mac mini contains 100 percent recycled zinc in the AC inlet prongs and multiple small parts.



Challenges

Technical properties

Availability and access

Traceability

Scale

Regulatory barriers

Supply chains